



Database Systems

(Course Code: **CSC270**)

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Theory Assignment# 1

Q1. Primary Keys and Constraints

Based on the schema description.

Primary Keys

Primary keys uniquely identify each record.

Relation	Primary Key
Doctors	DoctorID
Patients	PatientID
Appointments	AppointmentNo

Constraints

1. Entity Integrity

Primary key values cannot be **NULL**.

- Doctors.DoctorID → NOT NULL
- Patients.PatientID → NOT NULL
- Appointments.AppointmentNo → NOT NULL

2. Domain Constraints

Restrictions on attribute values.

Doctors

- DoctorID must start with '**D**'
- ConsultationFee must be **DECIMAL**
- All attributes are **mandatory (NOT NULL)**

Patients

- PatientID must start with '**P**'
- Age must be numeric
- Gender must be valid values (Male/Female etc.)

3. Referential Integrity (Foreign Keys)

Patients table

DoctorID references Doctors.DoctorID

Constraint:

- **ON UPDATE CASCADE**
- **ON DELETE CASCADE**

Meaning:

- If a doctor ID changes → automatically updated in Patients.
- If a doctor is deleted → related patient records are deleted.

Appointments table

- PatientID references Patients.PatientID
- DoctorID references Doctors.DoctorID

Constraint:

- **ON UPDATE SET NULL**
- **ON DELETE SET NULL**

Meaning:

- If doctor or patient changes → appointment values become NULL.
- If doctor or patient is deleted → appointment values become NULL.

Q2. Database Modification Analysis

a) Insert

(D005, Dr. Watson, Pediatrician, 120.00, 044-26578911)

Result: Executed Successfully

Reason:

- DoctorID starts with **D**
- Fee is decimal
- No PK conflict

b) Insert

(009, Dr. Henry, Dermatologist, 140.00, 078-225278911)

Result: Constraint Violation

Violation: Format restriction

Reason:

- DoctorID must start with **D**
- 009 does not satisfy the rule.

c) Insert Patient

(P00024, Michael, 28, Male, New York, Allergies, 2024-03-05, D005)

Result: Executed Successfully

Reason:

- PatientID starts with **P**
- DoctorID **D005 exists** (inserted earlier)
- No constraint violated.

d) Insert Patient

(P00025, Sophia, 35, Female, Chicago, Asthma, 2024-02-22, D010)

Result: Constraint Violation

Violation: Referential Integrity (Foreign Key)

Reason:

- DoctorID **D010 does not exist** in Doctors table.

e) Update DoctorID

D003 → D010

Result: Executed Successfully

Effect:

Due to **ON UPDATE CASCADE**

Patients table changes:

Patient	Old Doctor	New Doctor
P0003	D003	D010

Appointments table:

- DoctorID becomes **NULL** (ON UPDATE SET NULL)

f) Update PatientID

P0003 → P00026

Result: Executed Successfully

Effect in **Appointments**

- PatientID becomes **NULL** (ON UPDATE SET NULL)

g) Delete Doctor

DoctorID = D002

Result: Executed Successfully

Effects:

Patients table

- Patients referencing D002 are **deleted**
(ON DELETE CASCADE)

Patient deleted:

- **P0002**

Appointments table

DoctorID referencing D002 → **set to NULL**

Q3. Relational Algebra Queries

Relational Algebra Queries

The following relational algebra expressions are written over the database schema described in Question 1, using the instances from Question 2. Symbols used: σ (select), π (project), $\bowtie$ (natural join), γ (aggregate).

Part	Description	Relational Algebra Expression
a	Doctors with fee > \$150	$\sigma_{\{\text{ConsultationFee} > 150\}}(\text{Doctors})$
b	DoctorID & Name of Cardiologists	$\pi_{\{\text{DoctorID}, \text{Name}\}}(\sigma_{\{\text{Specialization} = \text{'Cardiologist'}\}}(\text{Doctors}))$
c	Patients with Age > 40	$\sigma_{\{\text{Age} > 40\}}(\text{Patients})$
d	Doctor name for patient P0002	$\pi_{\{\text{Name}\}}(\text{Doctors} \bowtie_{\{\text{Doctors.DoctorID} = \text{Patients.DoctorID}\}} \sigma_{\{\text{PatientID} = \text{'P0002'}\}}(\text{Patients}))$
e	PatientID, Name, City from Chicago, last visit after 2024-01-01	$\pi_{\{\text{PatientID}, \text{Name}, \text{City}\}}(\sigma_{\{\text{City} = \text{'Chicago'} \wedge \text{LastVisitDate} > \text{'2024-01-01'}\}}(\text{Patients}))$
f	Doctor & Patient name for Appointment 1002	$\pi_{\{\text{Doctors.Name}, \text{Patients.Name}\}}((\sigma_{\{\text{AppointmentNo} = 1002\}}(\text{Appointments})) \bowtie_{\{\text{Appointments.DoctorID} = \text{Doctors.DoctorID}\}} \text{Doctors} \bowtie_{\{\text{Appointments.PatientID} = \text{Patients.PatientID}\}} \text{Patients})$
g	Count of patients per doctor	$\gamma_{\{\text{DoctorID}, \text{COUNT(PatientID)}\}}(\text{Patients})$
h	Appointments between 01-01-2024 and 01-03-2024	$\sigma_{\{\text{AppointmentDate} \geq \text{'2024-01-01'} \wedge \text{AppointmentDate} \leq \text{'2024-03-01'}\}}(\text{Appointments})$